Anthony Baietto

anthony@baietto.dev | anthony.baietto.dev | Google Scholar
(815) 527-2452 | Linkedin | Blacklick, OH

EXPERIENCE

Artificial Intelligence Engineer **Jan 2025 – Present**
Applied Research Solutions *Beavercreek, OH*

- Conducted IR&D focusing on LLM applications for boosting personnel efficiency
- Constructed locally-hosted RAG/chatbot application for assisting day-to-day corporate task automation
- Managed team of software developers for synthetic data generation
- Led AI development for 2 Small Business Innovation Research (SBIR) Navy programs involving automation of RMF cybersecurity and system modeling processes

Artificial Intelligence Software Developer **Jan 2020 – Dec 2024**
Applied Research Solutions *Beavercreek, OH*

- Collaborated with Air Force Research Laboratory on radar waveform design (AutoWav project)
- Developed novel neural network solution for interference mitigation resulting in over $2000x$ speedup

Graduate Teaching Assistant **AU21, SP22, AU23, SP24, AU24**
The Ohio State University *Columbus, OH*

- Taught operating systems with responsibilities including lecturing and preparing assignments/exams
- Mean student evaluation score: 4.47 / 5.00 (department average: 4.26)
- Winner of Elanor Quinlan Graduate Teaching Award (2023)

Graduate Research Assistant **Aug 2020 – Dec 2024**
The Ohio State University *Columbus, OH*

- Developed innovative AI dataset augmentation techniques for neuromorphic computation
- Introduced neuromorphic computing obstacle along with generative AI mitigation

Undergraduate Research Assistant **Aug 2019 – Dec 2019**
ReRout Lab *Columbus, OH*

- Constructed demonstration of SoftwarePilot, a fully autonomous aerial system

IT Intern **May 2019 – Aug 2019**
CPTechnologies Company *Blacklick, OH*

- Developed and maintained real-time production management software with database support
- Networked and debugged 40+ kiosks and remote terminals
- Provided technical assistance for 30+ employees

EDUCATION

Ph.D. in Computer Science and Engineering **Jan 2020 – Dec 2024**
The Ohio State University *Columbus, OH*

- Thesis: Data-Aware Tuning of Deep Learning Models

M.S. in Computer Science and Engineering **May 2024**
The Ohio State University *Columbus, OH*

B.S. in Computer Science and Engineering **Aug 2017 – May 2020**
The Ohio State University *Columbus, OH*

- BS/MS Program, Maximus Scholarship, National Buckeye Scholar, Dean's List

SKILLS & INTERESTS

Technical Skills: AI/ML, LLM/RAG, LangChain, LlamaIndex, TensorFlow, PyTorch, Python, C/C++, Bash, REST
Skills: Complex Concept Translation (Technical & Non-Technical Audiences), Problem Solving, People & Project Leadership
Interests: Homelab, Fishing, Video Games, Star Wars, DIY Handyman

PUBLICATIONS & PRESENTATIONS

- **Baietto, A.**; Stewart, C.; Bihl, T.J. Dataset Assembly for Training Spiking Neural Networks. Neurocomputing 2025, 622, 129207. https://doi.org/10.1016/j.neucom.2024.129207
- **A. Baietto** and T. Bihl, "Generative Data for Neuromorphic Computing," 2025 Hawaii International Conference on System Sciences (HICSS), Big Island, HI, USA, 2025, pp. 7246-7255, https://hdl.handle.net/10125/109719
- **A. Baietto**, C. Stewart and T. Bihl, "Dataset Augmentation for Robust Spiking Neural Networks," 2023 IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C), Toronto, ON, Canada, 2023 pp. 116-121. doi: 10.1109/ACSOS-C58168.2023.00050
- Poster Presentation, "Toward Robust Spiking Neural Networks", International Conference on Neuromorphic Systems (ICONS) (2023)
- **A. Baietto**, J. Boubin, P. Farr and T. J. Bihl, "Lean Neural Networks for Real-time Embedded Spectral Notching Waveform Design," 2022 IEEE 31st International Symposium on Industrial Electronics (ISIE), Anchorage, AK, USA, 2022, pp. 1121-1126, doi: 10.1109/ISIE51582.2022.9831772.
- **Baietto, A.**; Boubin, J.; Farr, P.; Bihl, T.J.; Jones, A.M.; Stewart, C. Lean Neural Networks for Autonomous Radar Waveform Design. Sensors 2022, 22, 1317. https://doi.org/10.3390/s22041317

PATENTS

- **U.S. Patent Application No. 63/718,921** "SYSTEMS AND METHODS FOR TRAINING NEURAL NETWROKS," filed November 11, 2024. Patent pending.
- **U.S. Patent No. US 2025/0020775 A1** "METHOD OF ANALYZIING AND CORRECTING A COMPLEX WAVEFORM BY REAL AND IMAGINARY PARTITIONING AND RECOMBINATION," published January 16, 2025.
- **U.S. Patent No. US 2024/0249139 A1** "METHOD OF ANALYZING AND CORRECTING A DYNAMIC WAVEFORM USING MULTIVARIATE ERROR LOSS FUNCTIONS," published July 25, 2024.